

Problem 1) $J(\alpha) = \int_{x_1}^{x_2} f(y(x), y'(x); x) dx$

$$\Rightarrow \frac{\partial J}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} \frac{\partial y}{\partial \alpha} + \frac{\partial f}{\partial y'} \frac{\partial y'}{\partial \alpha} \right) dx ; \quad \frac{\partial y}{\partial \alpha} = \eta(x); \quad \frac{\partial y'}{\partial \alpha} = \frac{d\eta}{dx}$$

$$\Rightarrow \frac{\partial J}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} \eta(x) + \frac{\partial f}{\partial y'} \frac{d\eta(x)}{dx} \right) dx$$

Integration by parts gives us, $\int u dv = uv - \int v du$

$$\Rightarrow \int_{x_1}^{x_2} \frac{\partial f}{\partial y'} \frac{d\eta}{dx} dx = \underbrace{\frac{\partial f}{\partial y'} \eta(x)}_{=0 \text{ because } \eta(x_1) = \eta(x_2) = 0} \Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \eta(x) dx$$

$$\Rightarrow \frac{\partial J}{\partial \alpha} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} \eta(x) - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \eta(x) \right] dx$$

$$= \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right] \eta(x) dx$$

$$\frac{\partial J}{\partial \alpha} = 0 ; \eta(x) \text{ is arbitrary} \Rightarrow \boxed{\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right)} \text{ Euler's eqn.}$$

Problem 2) $L = T - U ; \quad \frac{\partial L}{\partial x_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} = 0 \quad i = 1, 2, 3$

$$\Rightarrow \frac{\partial (T - U)}{\partial x_i} - \frac{d}{dt} \frac{\partial (T - U)}{\partial \dot{x}_i} = 0 ; \text{ Recall that for conservative system,}$$

$$U \equiv U(x_i) ; T \equiv T(\dot{x}_i) \Rightarrow \frac{\partial U}{\partial x_i} = \frac{\partial T}{\partial \dot{x}_i} = 0$$

$$\Rightarrow \left. \begin{aligned} -\frac{\partial U}{\partial x_i} &= \frac{d}{dt} \frac{\partial T}{\partial \dot{x}_i} \\ \text{For conservative systems } F_i &= -\frac{\partial U}{\partial x_i} \end{aligned} \right\} \Rightarrow \frac{d}{dt} \frac{\partial T}{\partial \dot{x}_i} = \frac{d}{dt} \frac{\partial}{\partial \dot{x}_i} \left(\sum_j \frac{1}{2} m \dot{x}_j^2 \right)$$

$$= \frac{d}{dt} (m \dot{x}_i) = \dot{p}_i$$

$$= F_i \Rightarrow \boxed{F_i = \dot{p}_i}$$

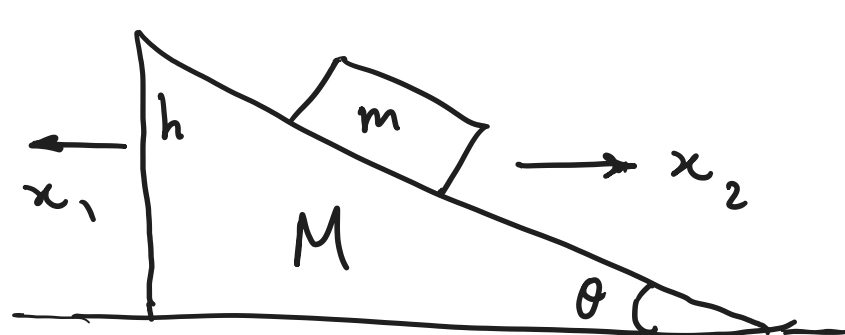
Problem 3) $L = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz ; \quad \frac{\partial L}{\partial x_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) = 0$

There is no x, y dependence; $\frac{\partial L}{\partial \dot{x}} = m\dot{x} \equiv \text{const}$; $\frac{\partial L}{\partial \dot{y}} = m\dot{y} \equiv \text{const}$.

Problem 4) x_1 : The horizontal coordinate of M .

x_2 : The horizontal coordinate of m .

$x_1 + x_2$: The relative horizontal distance between M, m .



$h = (x_1 + x_2) \tan \theta$ which is the height fallen by m .

$$\Rightarrow L = \frac{1}{2} M \dot{x}_1^2 + \frac{1}{2} m (\dot{x}_2^2 + \dot{h}^2) + mgh \quad (\text{The potential energy of } m \text{ is constant})$$

The E-L eqns for x_1, x_2 yield: $\begin{cases} M \ddot{x}_1 + m(\ddot{x}_1 + \ddot{x}_2) \tan^2 \theta = mg \tan \theta \\ m \ddot{x}_2 + m(\ddot{x}_1 + \ddot{x}_2) \tan^2 \theta = mg \tan \theta \end{cases}$

$$\Rightarrow \boxed{\ddot{x}_1 = \frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta}}$$

$$M \ddot{x}_1 - m \ddot{x}_2 = 0$$

$$\Rightarrow \frac{d}{dt} (M \dot{x}_1 - m \dot{x}_2) = 0$$

$\Rightarrow$ Linear momentum is constant